



Old Age Security: Calculation Methodology

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Agenda

- Executive Summary
- Background Information
- Calculations
- Data Analysis
- Recommendations





Overview

Purpose:

- To provide guidance on when clients should begin receiving OAS payments

Core Problem:

- Inconsistent advice on when clients should start OAS benefits



Solution:

- OAS Optimal Deferred Age Calculator
 - Determines the best age to receive payments
 - Gender, residency, and income as inputs



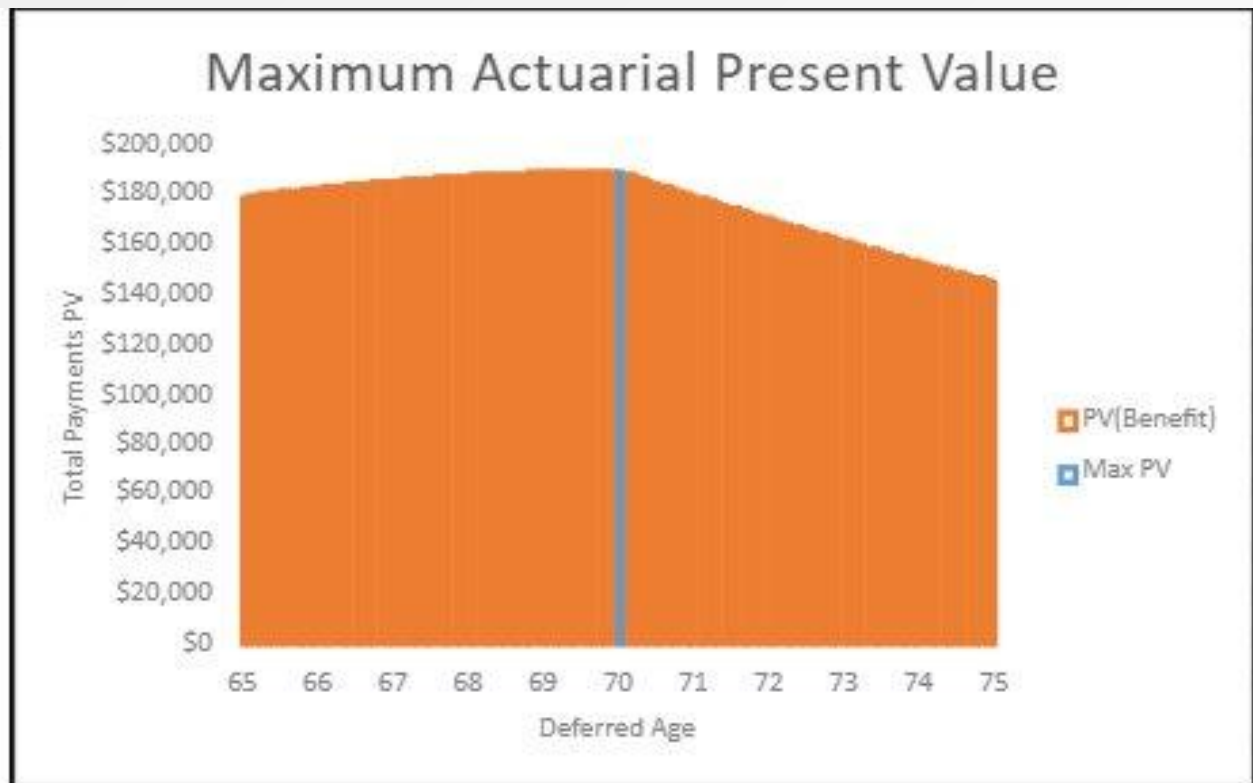


OAS Optimal Deferred Age Calculator

Client's Input:	
Yearly Income	\$ 137,000
Gender	Female
Residency from age 18 to age 65	40

Time Value:	
Interest Rate	3.0%

Optimal solution:	
Max PV	\$ 189,656
Age to receive payment	70
Monthly Payment:	
Before age 75	\$ 989.63
75 and above	\$ 1,088.59



You should defer your OAS payment to 0 months after age 70.





Key Insights

- Gender: Males receive payments earlier (shorter life expectancy)
- Residency: More years = higher payment value
- Income: Cannot receive payments if prior-year income exceeds threshold
- Interest Rates: Higher rates = lower payment value





Old Age Security

- A taxable monthly pension for Canadians seniors
- Provides fixed monthly payments
- From age 75 onwards, monthly payments increase 10%
- 0.6% payment increase on each deferral, up to age 70

Eligibility Rules:

- Age 65+
- Canadian citizen/legal resident
- 10+ years of residency since age 18
- 20+ years if living abroad
- Last year's annual net income does not exceed a stated threshold





Assumptions – Client Profile

- Annual Net Income:
 - Clients' annual net income is fixed
 - OAS deduction from recovery tax is not considered
- Gender:
 - Gender group affects the mortality rates
- Residency:
 - At least 10 years of residency in Canada after age 18
 - Benefits increase proportionally up to a maximum of 40 years
 - After age 65, client will not leave the country for a period of more than 6 months





Assumptions – Mortality Rate

Geography ⁵	Canada ⁶ (map)				
Sex	Both sexes				
Element	Death probability between age x and x+1 (qx)				
Age group	2019	2020	2021	2022	2023
	Number				
64 years	0.00805	0.00843	0.00861	0.00868	0.00834
65 years	0.00884	0.00923	0.00942	0.00950	0.00912
66 years	0.00972	0.01012	0.01031	0.01042	0.00999
67 years	0.01070	0.01111	0.01130	0.01144	0.01096
68 years	0.01179	0.01221	0.01239	0.01257	0.01203
69 years	0.01299	0.01344	0.01361	0.01382	0.01321
70 years	0.01433	0.01480	0.01496	0.01522	0.01454
⋮					
105 years	0.42264	0.44040	0.41933	0.44461	0.42786
106 years	0.44314	0.46063	0.43980	0.46469	0.44835
107 years	0.46274	0.47981	0.45941	0.48370	0.46791
108 years	0.48135	0.49786	0.47805	0.50155	0.48642
109 years	0.49888	0.51472	0.49564	0.51820	0.50380
110 years and over	1.00000	1.00000	1.00000	1.00000	1.00000

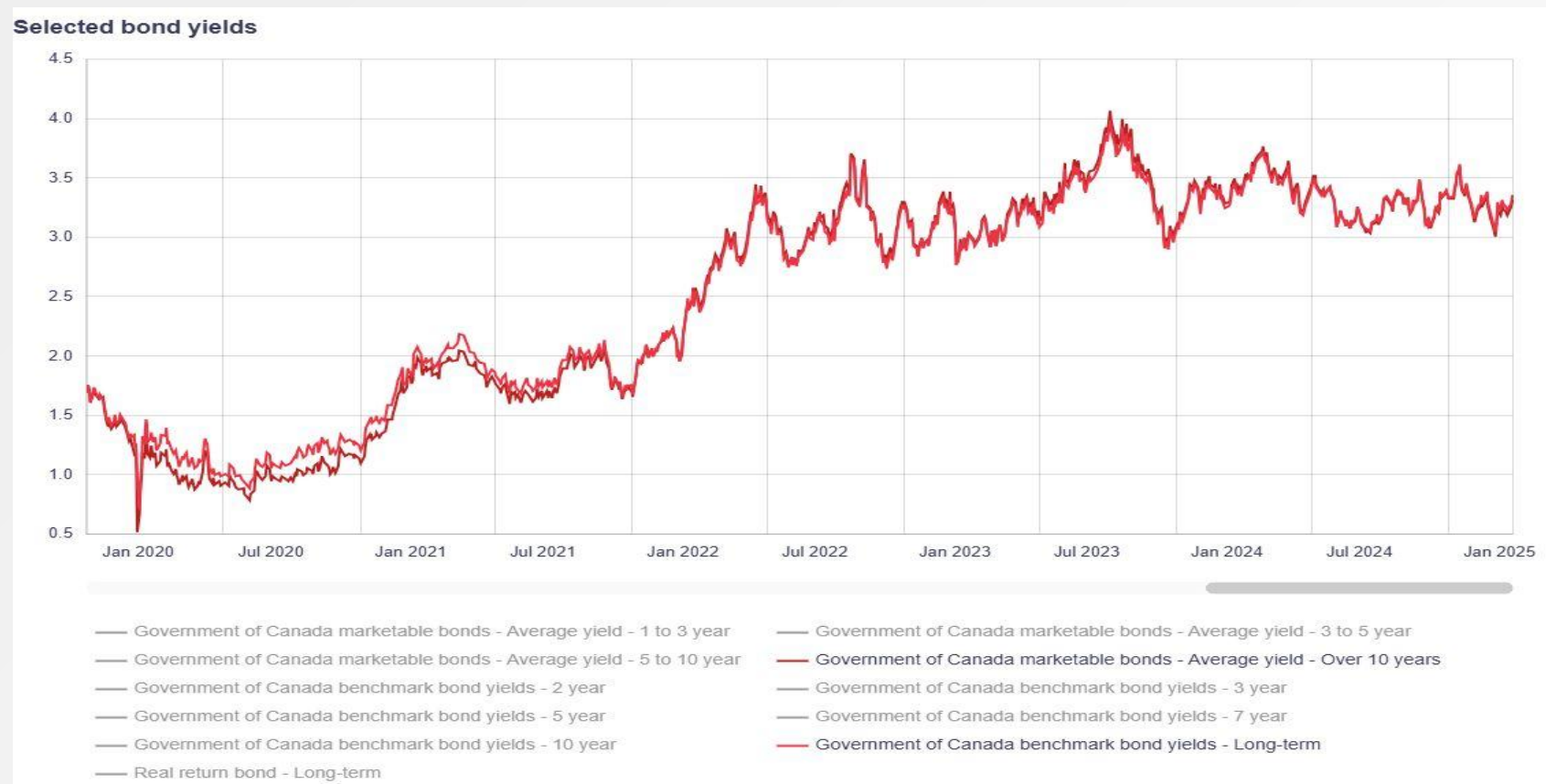




Assumptions – Interest Rate

Bank of Canada:

- Got 3% from long-term government bond rate



Deferred Actuarial Present Value (APV)

Total APV (u – month deferred after age 65)

$$= \sum_{k=0}^{540} \left[\text{Payment}_{u,k} \times \frac{\text{survival probability to age } (65 + k \text{ months}) \text{ given at age 65}}{(1 + \text{interest rate})^{\frac{k}{12}}} \right]$$

where

$$\text{Residency}_u = \left\lfloor \text{Residency at age 65} + \frac{u}{12} \right\rfloor$$

$$\text{Payment}_{u,k} = \begin{cases} \text{Payment}_u, & \text{if } u \leq k < 120, \text{Residency}_u \geq 10 \text{ and } \text{Income}_k \leq \$142,609 \\ 1.1 \times \text{Payment}_u, & \text{if } k \geq \max(u, 120), \text{Residency}_u \geq 10 \text{ and } \text{Income}_k \leq \$148,179 \\ 0, & \text{otherwise.} \end{cases}$$



Deferred Payment Amount



- Let $Deferral_u = \min(u, 60)$ be the deferral months subject to 0.6% benefit increase
- For more than 40 years of residence,

Full Deferred Payment:

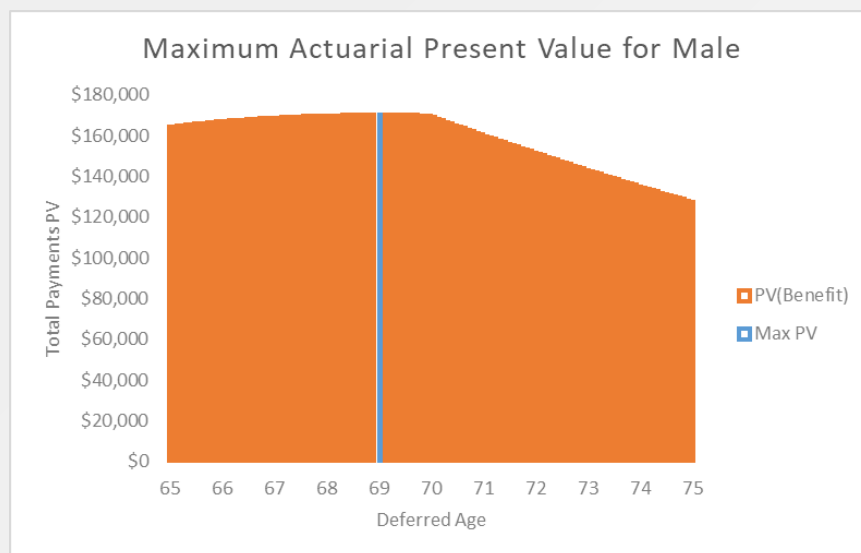
$$Payment_u = \$727.67 \times [1 + 0.6\%(Deferral_u)]$$



Maximum APV

- The Maximum APV among all $u = 0, 1, \dots, 540$ will yield the optimal deferral age $65 + \frac{u^*}{12}$

$$Total\ APV(u^*) = \max_{u=0,1,\dots,540} Total\ APV(u)$$

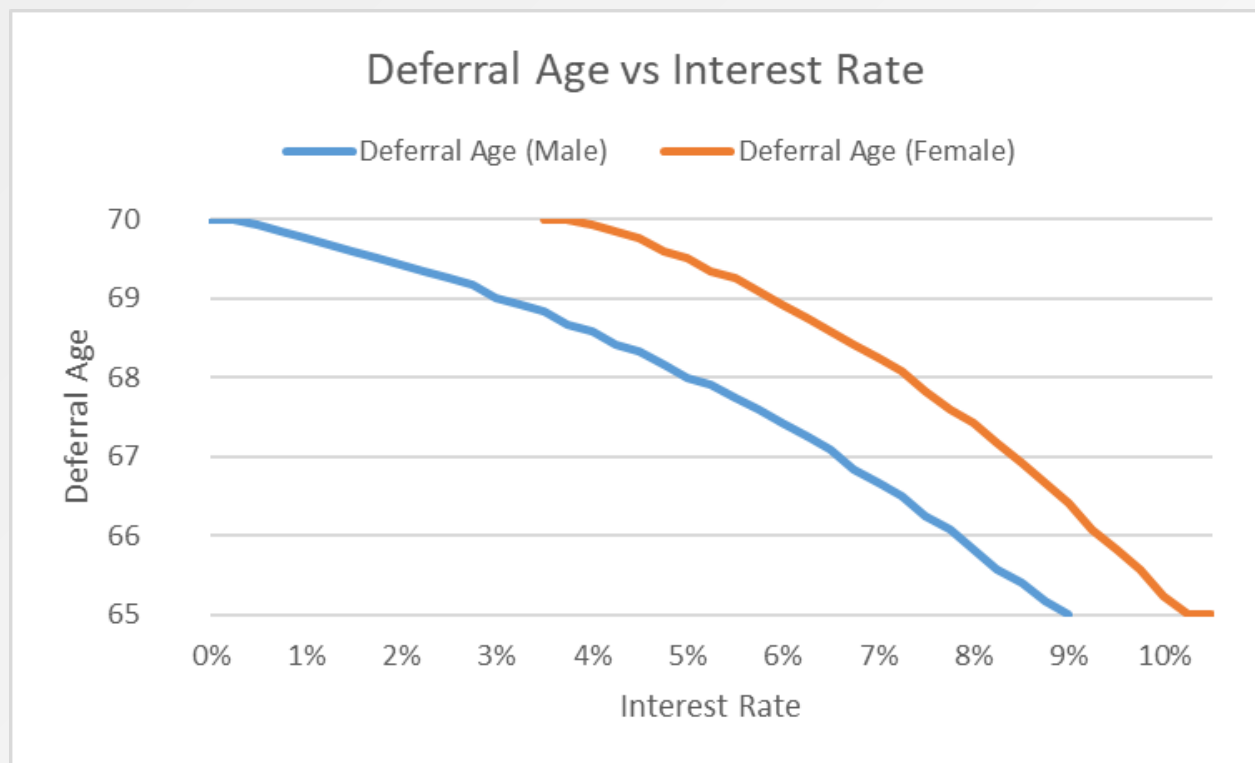


Client's Input:	By Gender		
Income	\$ 100,000	\$ 100,000	\$ 100,000
Gender	Female	Unspecified	Male
Residency	40	40	40
Interest Rate	3.00%	3.00%	3.00%
Optimal solution:			
Max PV	\$ 189,656	\$ 180,447	\$ 170,169
Deferral Age	70	69 9/12	69
Monthly Payment:			
Before age 75	\$ 989.63	\$ 976.53	\$ 937.24
75 and above	\$ 1,088.59	\$ 1,074.19	\$ 1,030.96

Higher the Mortality Rate, Earlier the Deferral Is.

Interest Rate

- Higher the Interest Rate, Earlier the Deferral is.





Years of Residence

- Years of Residence is counted from age 18.
- Benefit increases proportionally with years of residency up to a maximum of 40 years.
- If client is not eligible within age 75, the optimal solution is as soon as eligible.
- Any periods with more than 6 months outside Canada are not counted.



Partial Payment Amount

Recall: $Residency_u = \left\lfloor Residency \text{ at age } 65 + \frac{u}{12} \right\rfloor$

There are two formulas for computing the payment amounts:

- Deferral-dominant:

$$Payment_u^d = \$727.67 \times \left\lfloor \frac{\min(Residency_0, 40)}{40} \right\rfloor \times [1 + 0.6\%(Deferral_u)]$$

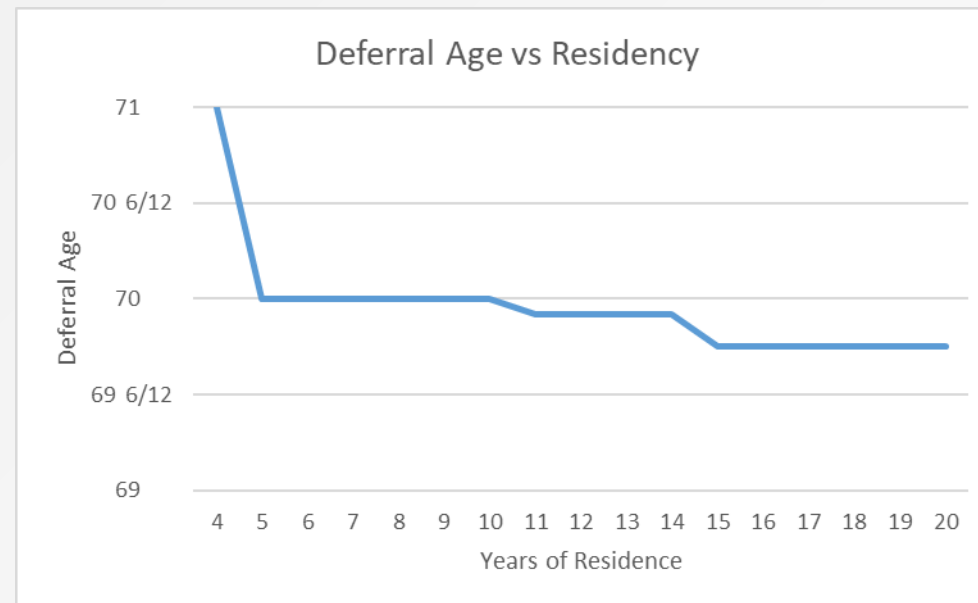
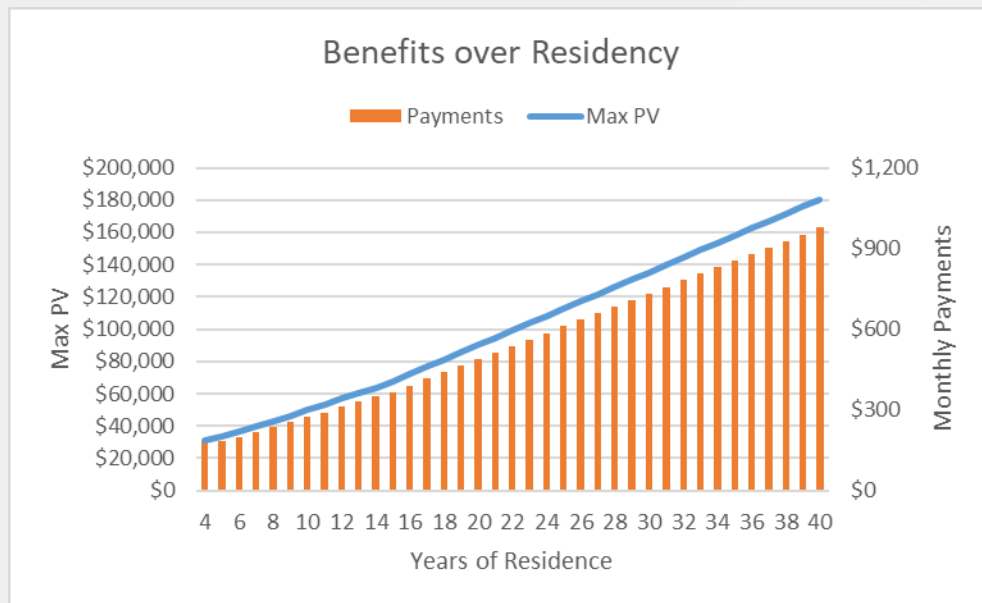
- Residency-dominant:

$$Payment_u^r = \$727.67 \times \left\lfloor \frac{\min(Residency_u, 40)}{40} \right\rfloor \times [1 + 0.6\%(Deferral_u \bmod 12)]$$

Therefore, the actual u-month deferred payment after age 65 is:

$$Payment_u = \max(Payment_u^d, Payment_u^r)$$

Residency

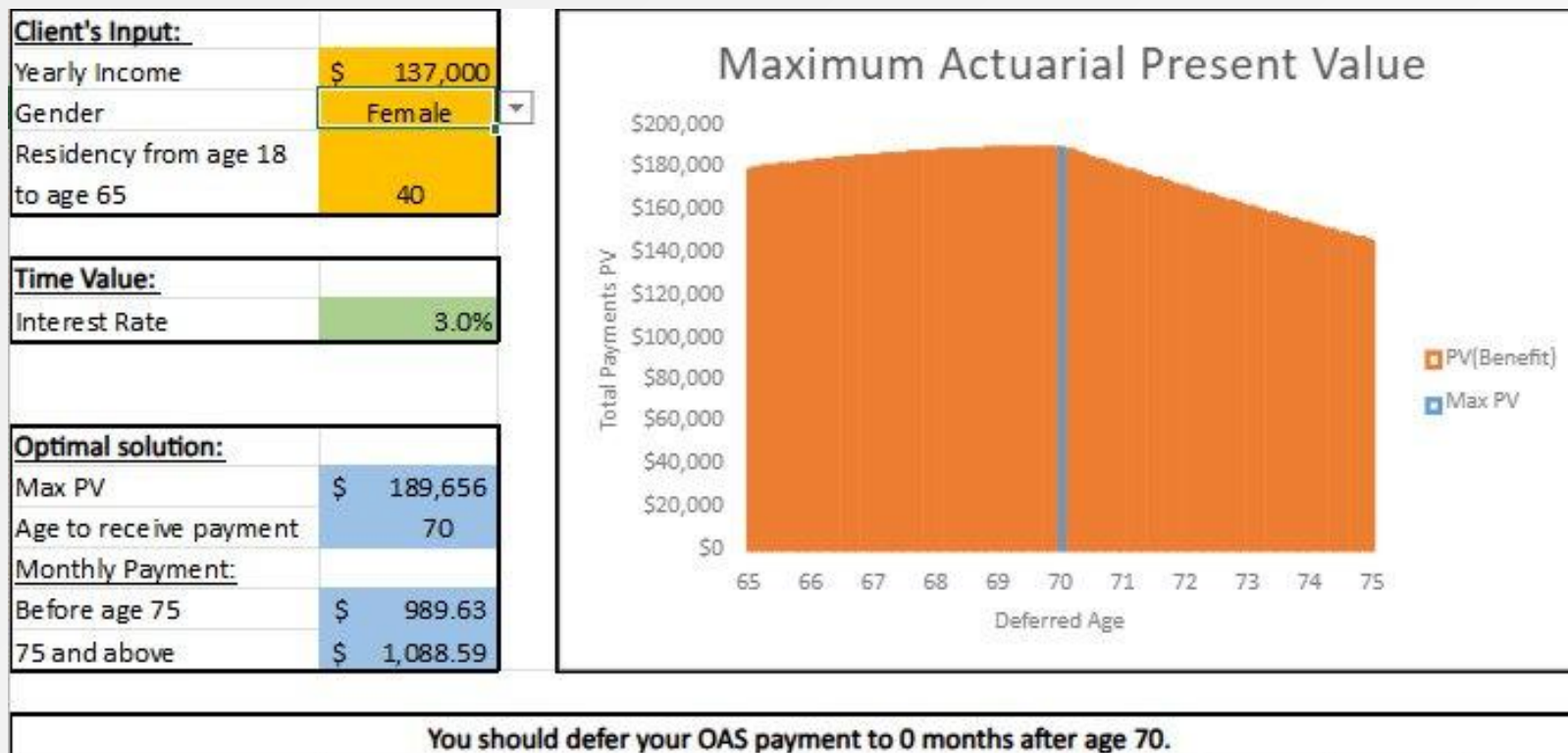


- Residency at age 65 less than 15 years suggests a further deferral.



Recommendations

- Use of the OAS Optimal Deferred Age Calculator





Thank you!

Feel free to ask questions.